

How to give read and write permissions using Unix commands.
A very simple example is to create a text file with a C program,
then use `chmod`.

Create a file called `permission.c`

```
#include <stdio.h>
int main()
{
    FILE *fp;
    fp = fopen("sample.txt", "w");
    if (fp == NULL)
    {
        printf("Cannot open file\n");
        return 1;
    }
    fprintf(fp, "Hello, this is a test file.\n");
    fclose(fp);
    printf("File created successfully.\n");
    return 0;
}
```

Compile

```
gcc permission.c -o permission
```

Run

```
./permission
```

Output

File created successfully.

A file called sample.txt is created.

Check the current permissions

```
ls -l sample.txt
```

You may see

```
-rw-r--r-- 1 user user 31 Aug 17 08:20 sample.txt
```

The owner has: Read + Write

Group has: Read

Others have: Read

This is permission 644.

Give Read and Write permission to everyone

```
chmod 666 sample.txt
```

Check `ls -l sample.txt`

You should see

```
-rw-rw-rw- 1 user user ... sample.txt
```

Instead of numbers, you can directly give:

```
chmod u+rw sample.txt
```

```
chmod o+rw sample.txt
```

```
chmod a+rw sample.txt
```

Task of the Day

- 1) Give Read and Write only to the owner
- 2) Give Read and Write to the owner and group
- 3) Only owner can read and write
- 4) Owner read/write, everyone else read-only
- 5) Owner can do everything, others can only read and execute
- 6) Owner and group can read/write
- 7) Only owner can execute

Fill in the permission numbers for the following:

- 8) Owner: read + write; Group: nothing; Others: nothing
→ **600** _____
- 9) Owner: read + write; Group: read; Others: read → **644**

- 10) Owner: read + write + execute; Group: read + execute;
Others: read + execute → **755** _____
- 11) Owner: read + write; Group: read + write; Others:
nothing → **660** _____
- 12) Everyone: read only → **444** _____

```
Last login: Mon Aug 17 18:28:22 on ttys000
/var/folders/g6/zh7p_f2x2qbcf0_rwd7zsy08080gn/T/orbstack-open-terminal_0C70DAF8.sh ; exit;
harsha@Harsha-MacBook-Air3 ~ % /var/folders/g6/zh7p_f2x2qbcf0_rwd7zsy08080gn/T/orbstack-open-terminal_0C70DAF8.sh ; exit;
```

```
OrbStack update available!
Run "orb update" to update.

Updates include improvements, features, and fixes.
```

```
harsha@Harsha:~$ nano permission.c
harsha@Harsha:~$ ./permission
-bash: ./permission: No such file or directory
harsha@Harsha:~$ nano permission.c
harsha@Harsha:~$ gcc permission.c -o permission
harsha@Harsha:~$ ./permission
File created successfully.
harsha@Harsha:~$ ls -l sample.txt
-rw-r--r-- 1 harsha harsha 28 Aug 17 18:21 sample.txt
harsha@Harsha:~$ chmod 666 sample.txt
harsha@Harsha:~$ chmod 600 sample.txt
harsha@Harsha:~$ ls -l sample.txt
-rw----- 1 harsha harsha 28 Aug 17 18:21 sample.txt
harsha@Harsha:~$ chmod 666 sample.txt
harsha@Harsha:~$
harsha@Harsha:~$ ls -l sample.txt
-rw-rw-rw- 1 harsha harsha 28 Aug 17 18:21 sample.txt
harsha@Harsha:~$ chmod <number> sample.txt
ls -l sample.txt
-bash: number: No such file or directory
-rw-rw-rw- 1 harsha harsha 28 Aug 17 18:21 sample.txt
harsha@Harsha:~$ chmod 600 sample.txt
harsha@Harsha:~$ ls -l sample.txt
-rw----- 1 harsha harsha 28 Aug 17 18:21 sample.txt
harsha@Harsha:~$ chmod 666 sample.txt
harsha@Harsha:~$ ls -l sample.txt
-rw-rw-rw- 1 harsha harsha 28 Aug 17 18:21 sample.txt
harsha@Harsha:~$ chmod 600 sample.txt
harsha@Harsha:~$ ls -l sample.txt
-rw----- 1 harsha harsha 28 Aug 17 18:21 sample.txt
harsha@Harsha:~$ chmod 644 sample.txt
harsha@Harsha:~$ ls -l sample.txt
-rw-r--r-- 1 harsha harsha 28 Aug 17 18:21 sample.txt
harsha@Harsha:~$ chmod 755 sample.txt
harsha@Harsha:~$ ls -l sample.txt
-rwxr-xr-x 1 harsha harsha 28 Aug 17 18:21 sample.txt
harsha@Harsha:~$ chmod 666 sample.txt
harsha@Harsha:~$ ls -l sample.txt
-rw-rw-rw- 1 harsha harsha 28 Aug 17 18:21 sample.txt
harsha@Harsha:~$ chmod 100 sample.txt
harsha@Harsha:~$ ls -l sample.txt
ls: cannot access 'sample.txt': No such file or directory
harsha@Harsha:~$ chmod 100 sample.txt
harsha@Harsha:~$ ls -l sample.txt
--x----- 1 harsha harsha 28 Aug 17 18:21 sample.txt
harsha@Harsha:~$
```